IEOR 221

Homework 2

Module 3 (September 15, 2025)

1) A 1-year long forward contract on a non-dividend-paying stock is entered into when the stock price is $40 and the risk-free rate of interest is 5% per annum with continuous compounding.
(a) What are the forward price and the initial value of the forward contract?
(b) Six months later, the price of the stock is $45 and the risk-free interest rate is still 5%. What are the forward price and the value of the forward contract? (Hull 5.17)

2) Repeat the problem 1) if the stock pays cash dividend of $3 at the end of month 3 and month 9 (i.e., t = 0.25 and 0.75) and everything else are the same.

3) Assume that the risk-free interest rate is 4% per annum with continuous compounding and that the dividend yield on a stock index varies throughout the year. In February, May, August, and November, dividends are paid at a rate of 5% per annum. In other months, dividends are paid at a rate of 2% per annum. Suppose that the value of the index on July 31 is 1,300. What is the futures price for a contract deliverable in December 31 of the same year? (Hull 5.19)

4) Suppose that the risk-free interest rate is 6% per annum with continuous compounding and that the dividend yield on a stock index is 4% per annum. The index is standing at 400, and the futures price for a contract deliverable in four months is 405. What arbitrage opportunities does this create? (Hull 5.20)

5) The spot exchange rate between the Swiss franc and U.S. dollar is 1.0404 ($ per franc). Interest rates in the United States and Switzerland are 0.25% and 0% per annum, respectively, with continuous compounding. The 3-month forward exchange rate was 1.0300 ($ per franc). What arbitrage strategy was possible? How does your answer change if the forward exchange rate is 1.0500 ($ per franc). (Hull 5.32)